

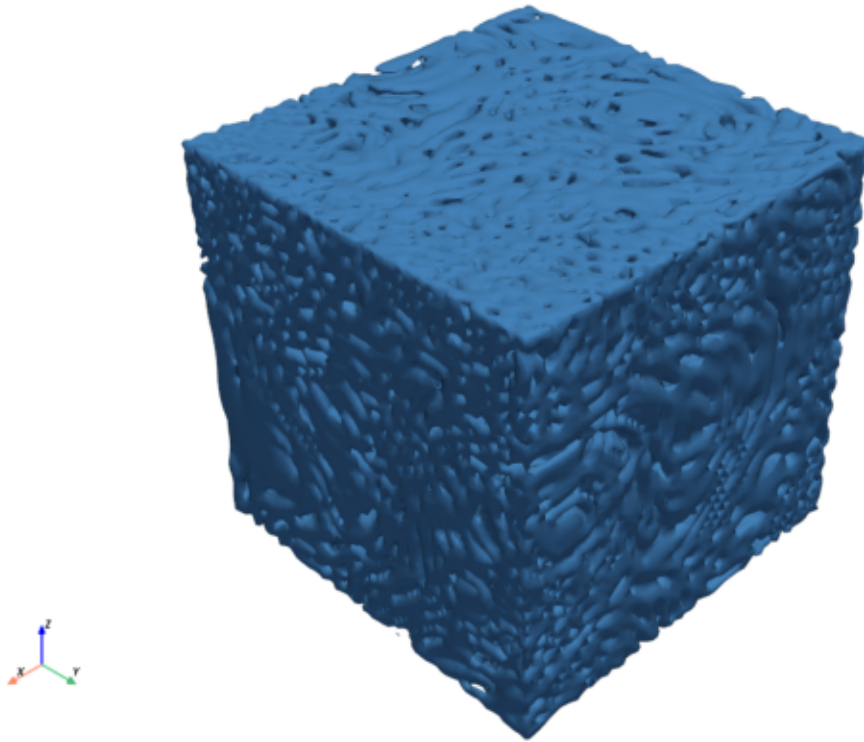
Bio-Inspired Heat Exchanger — Design Summary

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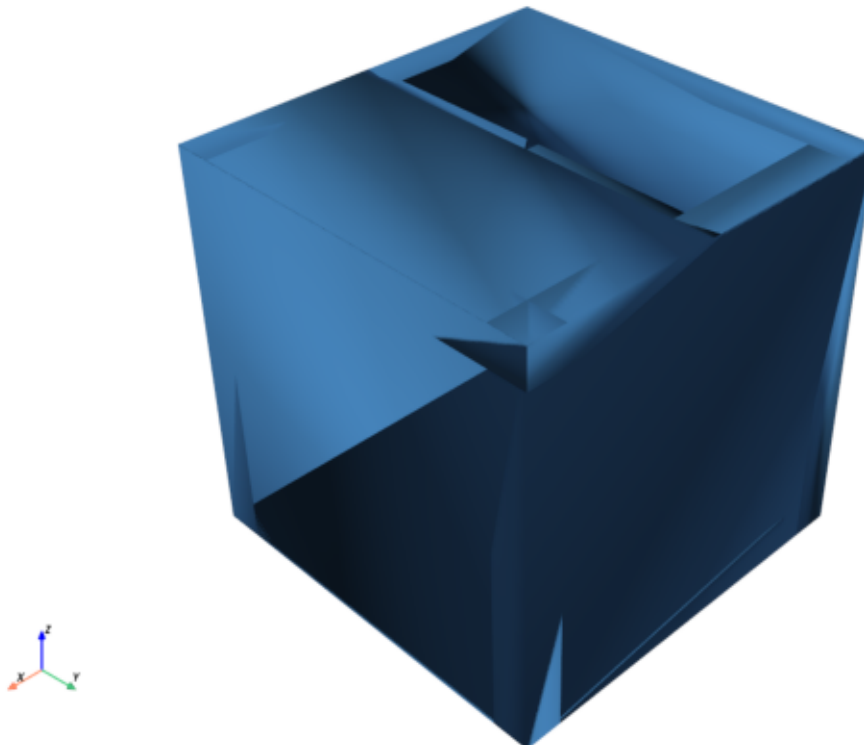
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encapsulation:
  length_mm..... 25.0
  width_mm..... 25.0
  height_mm..... 25.0
  wall_mm..... 3.0
  Dh_mm..... 19.0
  V_int_cm3..... 6.859
flow:
  v_in_m_s..... 0.1
  T_in_K..... 380.0
  p_out_Pa..... 0.0
  Re..... 126.4
  Pr..... 0.708
sizing:
  Re_simulation..... 126.4
  Re_operating..... 1000.0
  Nu_laminar..... 6.653
  Nu_turbulent..... 18.713
  target_Nu..... 10.0
  required_solidity..... 0.95
  wall_thickness_est_mm..... 5.7455
  h_W_m2K..... 25.31
  lam_turb_factor..... 2.8129
lattice:
  tpms..... gyroid
  k0_rad_m..... 628.0
  uc_mm..... 10.01
  phi..... 0.95
preview_geometry:
  solidity..... 0.1952
  mass_g..... 5.93
  surface_cm2..... 42.0
  n_triangles..... 126849
manufacturability:
  overhang_penalty..... 0.1157
  thickness_penalty..... 0.0
  combined_penalty..... 0.1157
  est_wall_thickness_mm..... 0.6218
  min_feature_size_mm..... 0.2
  max_overhang_angle_deg..... 45.0
  printable..... False
```

Optimised Geometry

gyroid_surface_lattice.stl



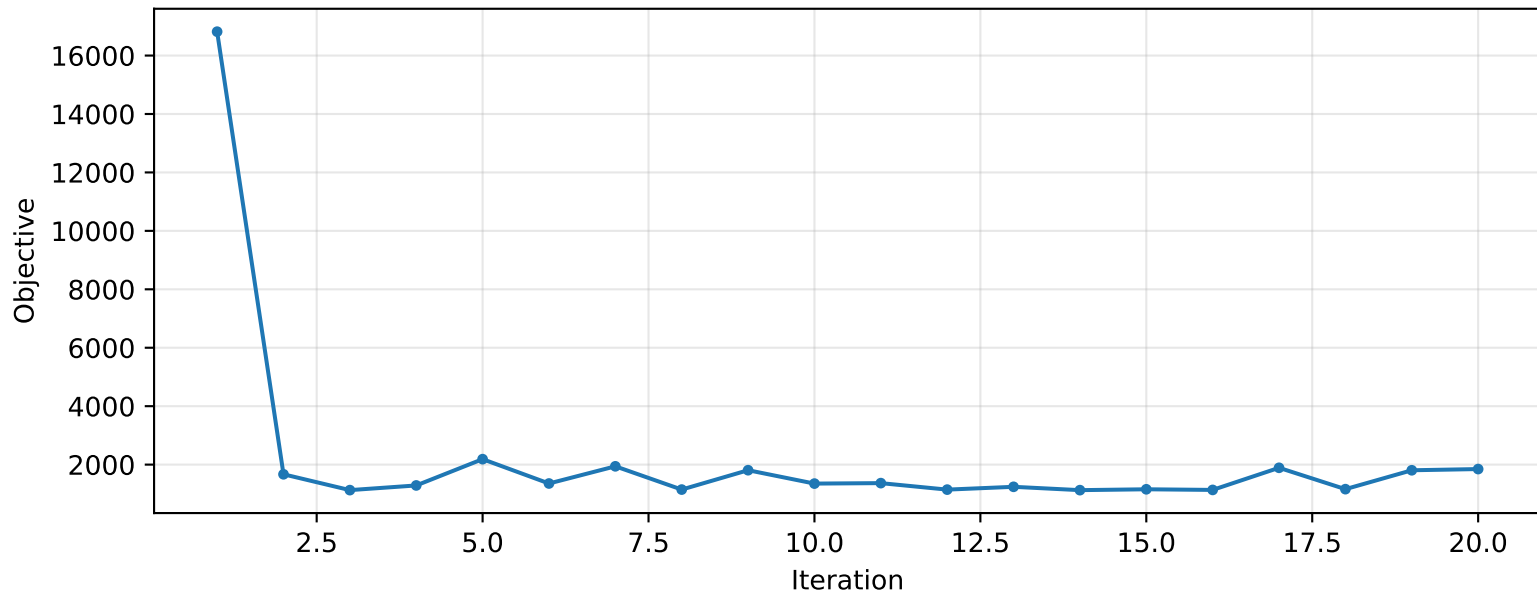
gyroid_surface_encap.stl



Optimisation History

iter	mode	J_meanT	J_obj	J_aug	DissPower	outletT	solid_frac	solid_g	flow_m3s	deltaP_m2s2	wall_thickness	g_meanT	pen_meanT	g_disspower	pen_disspower	mu_penalty	g_oh	pen_oh	mu_oh	grad_norm	delta_x	elapsed_s	alphaMax
1		0.101	0.101	1741	1.682e+04	270	0.7074	30.13	1e-07	168.2	1	nan	0	1.682e+04	1741	1e+04	0.002755	0.0536	8100	8.579e+05	nan	7.2	1.6e+07
2		0.068	0.068	0.135	1669	270	0.7122	30.34	1e-07	16.7	1	nan	0	1669	0	1.1e+04	0.002843	0.06698	8100	9.967	7.746	7.6	1.6e+07
3		0.376	0.376	0.4686	1127	270	0.7112	30.29	1e-07	11.3	1	nan	0	1127	0	1.1e+04	0.003312	0.0926	8100	32.83	7.746	7.8	1.6e+07
4		0.305	0.305	0.3979	1287	270	0.7081	30.16	1e-07	12.89	1	nan	0	1287	0	1.1e+04	0.00335	0.09292	8100	23.81	6.841	9.2	1.6e+07
5		0.41	0.41	0.5059	2187	270	0.706	30.07	1e-07	21.87	1	nan	0	2187	0	1.1e+04	0.003426	0.09586	8100	17.51	7.255	7.8	1.6e+07
6		0.411	0.411	0.5185	1354	270	0.7057	30.06	1e-07	13.54	1	nan	0	1354	0	1.1e+04	0.003533	0.1075	8100	20.07	6.757	11.6	1.6e+07
7		0.647	0.647	0.7831	1942	270	0.708	30.16	1e-07	19.44	1	nan	0	1942	0	1.1e+04	0.003762	0.1361	8100	14.2	6.715	9	1.6e+07
8		0.657	0.657	0.8063	1146	270	0.7088	30.19	1e-07	11.46	1	nan	0	1146	0	1.1e+04	0.003524	0.1493	8100	46.31	6.553	10.5	1.6e+07
9		0.491	0.491	0.5964	1809	270	0.707	30.11	1e-07	18.09	1	nan	0	1809	0	1.1e+04	0.003326	0.1054	8100	29.69	6.658	8.8	1.6e+07
10		0.593	0.593	0.7504	1350	270	0.7088	30.19	1e-07	13.52	1	nan	0	1350	0	1.1e+04	0.003986	0.1574	8100	34.18	6.391	8.9	1.6e+07
11		1.037	1.037	1.209	1366	270	0.7058	30.07	1e-07	13.68	1	nan	0	1366	0	1.1e+04	0.003773	0.1723	8100	49.6	6.051	9.6	1.6e+07
12		0.872	0.872	1.099	1144	270	0.703	29.95	1e-07	11.45	1	nan	0	1144	0	1.1e+04	0.004283	0.2266	8100	37.01	5.512	8.3	1.6e+07
13		0.846	0.846	1.026	1240	270	0.7095	30.22	1e-07	12.42	1	nan	0	1240	0	1.1e+04	0.003929	0.1797	8100	14.36	5.649	16.5	1.6e+07
14		1.065	1.065	1.267	1126	270	0.7038	29.98	1e-07	11.27	1	nan	0	1126	0	1.1e+04	0.004058	0.2019	8100	25.47	5.414	11.7	1.6e+07
15		0.992	0.992	1.324	1156	270	0.7006	29.84	1e-07	11.57	1	nan	0	1156	0	1.1e+04	0.004038	0.3319	8100	26.51	4.948	11.6	1.6e+07
16		1.181	1.181	1.345	1134	270	0.7047	30.02	1e-07	11.37	1	nan	0	1134	0	1.1e+04	0.003753	0.1643	8100	25.67	5.091	11.5	1.6e+07
17		1.207	1.207	1.44	1891	270	0.7016	29.89	1e-07	18.91	1	nan	0	1891	0	1.1e+04	0.003921	0.2326	8100	36.45	4.716	10.2	1.6e+07
18		1.172	1.172	1.351	1160	270	0.7011	29.87	1e-07	11.61	1	nan	0	1160	0	1.1e+04	0.003889	0.1789	8100	21.57	4.63	11.6	1.6e+07
19		1.093	1.093	1.27	1806	270	0.7024	29.92	1e-07	18.08	1	nan	0	1806	0	1.1e+04	0.003889	0.1769	8100	28.71	4.509	9.7	1.6e+07
20		1.081	1.081	1.243	1848	270	0.7016	29.89	1e-07	18.49	1	nan	0	1848	0	1.1e+04	0.003594	0.1623	8100	45.34	4.471	14.8	1.6e+07

Objective Convergence



Constraint Convergence

